EMETIC AND ANTIEMETIC DRUGS

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Contents

- Factors controlling vomiting
- Reflex mechanism of vomiting
- Emetics and causes of vomiting
- Antiemetic drugs
- Combination of antiemetic drugs

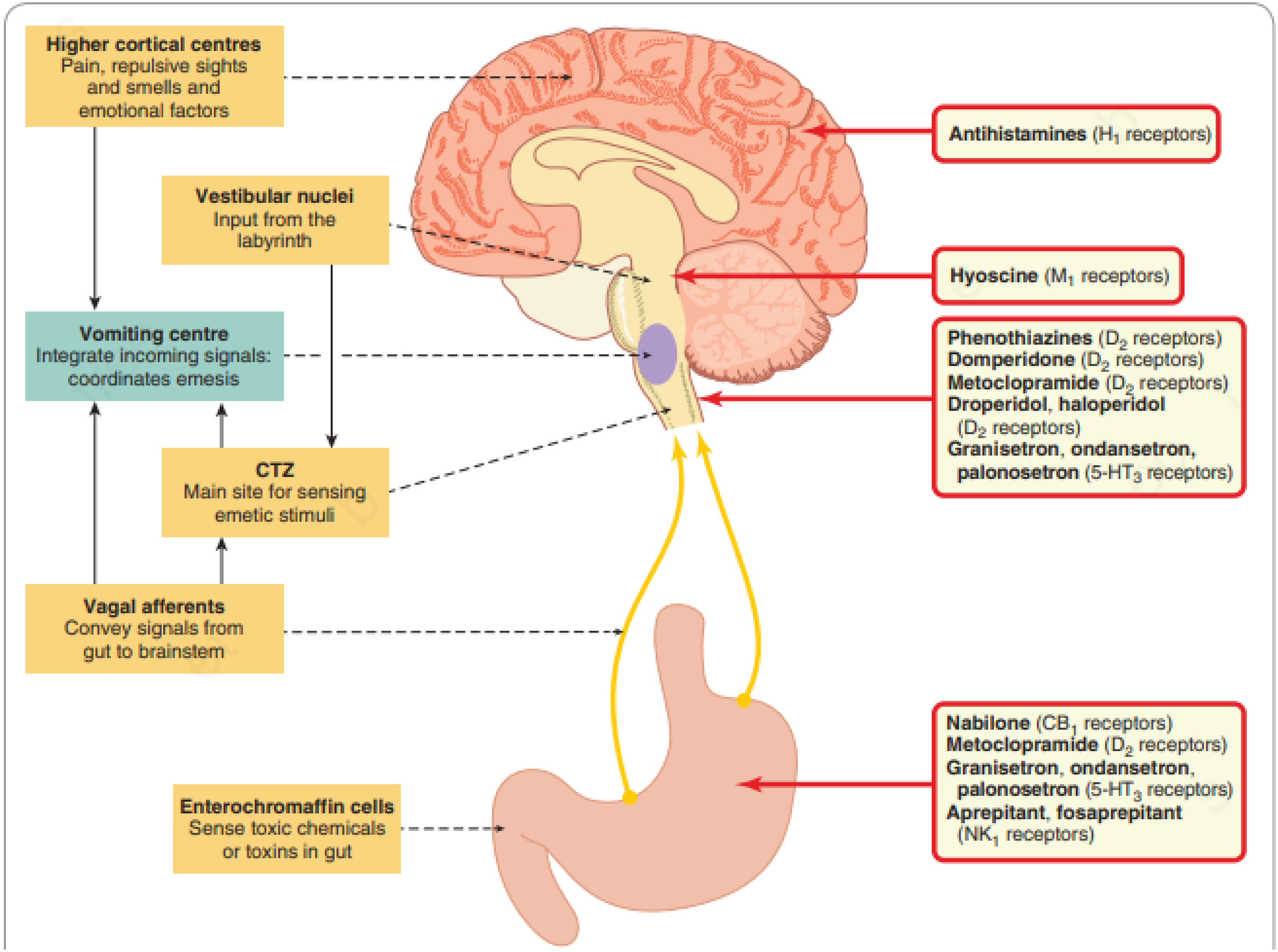
Vomiting or Emesis

Is an act of forceful expulsion of gastric contents through the mouth and it is often preceded by nausea

Causes of vomiting; gastritis, peptic ulcer. Intestinal obstruction, Drug induced, pregnancy. Sever pain, Brain tumor, Iatrogenic, ……

Neurotransmitters Involved

- Serotonin via 5HT3 receptors
- Acetylcholine via M receptors
- Dopamine via D2 receptors
- Histamine via H1 receptors



Schematic diagram of the factors involved in the control of vomiting, with the probable sites of action of antiemetic

Emetic Drugs

Apomorphine

Semi synthetic derivative of morphine
Given IM or SC , act centrally
Dose is 6 mg (2-8 mg)
Induces vomiting in 5-10 minutes
It has CNS depressant effect so it is contraindicated in Respiratoy failure

Ipecacuanha

Contains tow alkaloids EMETINE and CEPHALINE

Used as syrup Ipecae

Produces vomiting in 15 minutes

Acts by irritating gastric mucosa and CTZ center

Dose: 5ml in infants, 10-15 ml in children and 20 ml in adults

Contraindications:

Kerosene and corrosive poisoning

unconcious patients

Morphine poisoning

Antiemetic Drugs

RECEPTOR ANTAGONISTS

5-HT3 receptor antagonists: Granisetron, ondansetron, Dolasetron and palonosetron are of particular value in preventing and treating the vomiting and, to a lesser extent the nausea, commonly encountered **postoperatively** as well as that caused by **radiation therapy** or administration of **cytotoxic drugs** such as cisplatin. The primary site of action of these drugs is the CTZ. They may be given orally or by injection (sometimes helpful if nausea is already present). Excreted by kidney and liver, **no need dose reduction in renal impairment but yes in liver impairment.** Tissue half life 4-9 hrs while for Palonsetron is 40hrs.

Unwanted effects such as headache and GI upsets specially constipation and QT interval prolongation (more with Dolasetron) are relatively uncommon

Substituted Benzamide

Metoclopramide is a **D2 receptor antagonist**, closely related to the phenothiazine group, that acts **centrally on the CTZ** and also has a **peripheral action on the GI tract itself**, increasing the motility of the oesophagus, stomach and intestine make it useful in the treatment of **gastro-oesophageal reflux and hepatic and biliary disorders**.

Dose **10-20mg** every **6 hrs** orally or parentrally

Rapidly and completely absorbed after oral ingestion

Undergoes a high degree first pass metabolism

It is excreted in the urine as free and as metabolites and it also excreted in the breast milk

Unwanted effects of Metoclopramide

As metoclopramide also blocks dopamine receptors elsewhere in the central nervous system, it produces a number of unwanted effects including disorders of movement (more common in children and young adults), fatigue, motor restlessness, spasmodic torticollis (involuntary twisting of the neck) and oculogyric crises (involuntary upward eye movements). It stimulates prolactin release causing galactorrhoea, disorders of menstruation, impotence and gynecomastia.

Domperidone

It is a similar drug used to treat vomiting due to cytotoxic therapy as well as GI symptoms.

Unlike metoclopramide, it does not readily penetrate the blood– brain barrier and is consequently less prone to producing central side effects.

Domperidone is associated with a small increased risk of serious cardiac adverse effects (**particularly at higher doses and in older patients**), and its use is now restricted.

This drug is given orally, have plasma half-lives of 4–5 h and excreted in the urine.

Dopamine antagonists

Antipsychotic phenothiazines, such as chlorpromazine, perphenazine, prochlorperazine and trifluoperazine, are effective antiemetics.

Used for more severe nausea and vomiting associated with **cancer, radiation therapy, cytotoxic drugs, opioids, anaesthetics and other drugs**.

They can be administered orally, intravenously or by suppository.

They act mainly as **antagonists of the dopamine D2 receptors in the CTZ** but they also block **histamine and muscarinic receptors**.

Unwanted effects are common and include sedation (especially chlorpromazine), hypotension and extrapyramidal symptoms including dystonias and tardive dyskinesia.

Haloperidol, the related compound **droperidol and levomepromazine**, also act as D2 antagonists in the CTZ and can be used for acute chemotherapy-induced emesis.

NK1 receptor antagonists

Substance P causes vomiting when injected intravenously and is released by GI vagal afferent nerves as well as in the vomiting center itself.

Aprepitant, Rolapitant, Fosaprepitant and Netupitant block substance P (NK1) receptors in the CTZ and vomiting center.

Aprepitant is given orally, and is effective in controlling the late phase of emesis caused by cytotoxic drugs.

Aprepitant and Rolapitant undergo hepatic metabolism, primarily by CYP3A4.

Coadministration with strong inhibitors or inducers of CYP3A4 (for example, clarithromycin and St. John's wort herbal medicine, respectively) should be avoided.

Aprepitant is an inducer of CYP3A4 and CYP2C9, and it also exhibits dose dependent inhibition of CYP3A4. Therefore, it may affect the metabolism of other drugs that are substrates of these isoenzymes and is subject to numerous drug interactions.

Rolapitant is a moderate inhibitor of CYP2D6.

Fosaprepitant is a prodrug of aprepitant, which is administered intravenously

Fatigue, diarrhea, abdominal pain, and hiccups are adverse effects of this class.

H1 receptor antagonists

Cinnarizine, cyclizine and promethazine are the most commonly employed; they are effective against nausea and vomiting arising from many causes, including motion sickness and the presence of irritants in the stomach. None is very effective against substances that act directly on the CTZ. Promethazine is used for morning sickness of pregnancy (on the rare occasions when this is so severe that drug treatment is justified), and has been used by NASA to treat space motion sickness. Drowsiness and sedation

Muscarinic receptor antagonists

Hyoscine (scopolamine)

Is employed principally for prophylaxis and treatment of motion sickness, and may be administered orally or as a transdermal patch.

Dry mouth and blurred vision are the most common unwanted effects.

Drowsiness also occurs, but the drug has less sedative action than the antihistamines because of poor central nervous system penetration

OTHER ANTIEMETIC DRUGS

Cannabinoids

The synthetic cannabinol Nabilone decreases vomiting caused by agents that stimulate the CTZ, and is sometimes effective where other drugs have failed.

Nabilone is given orally; it is well absorbed from the GI tract and is metabolised in many tissues with a **plasma half-life is approximately 120 min,** and its metabolites are excreted in **the urine and faeces**.

Unwanted effects are common, especially **drowsiness, dizziness and dry mouth,** mood changes and postural hypotension.

The antiemetic effect is antagonised by naloxone, which implies that opioid receptors may be important in the mechanism of action.

High-dose glucocorticoids (particularly dexamethasone; can also control emesis, especially when this is caused by cytotoxic drugs. The mechanism of action is not clear, but it may involve blockade of prostaglandins..

Dexamethasone is typically deployed in combination with metoclopramide or ondansetron in patients receiving cytotoxics, or in postoperative nausea and vomiting.

Benzodiazepines

Alprazolam

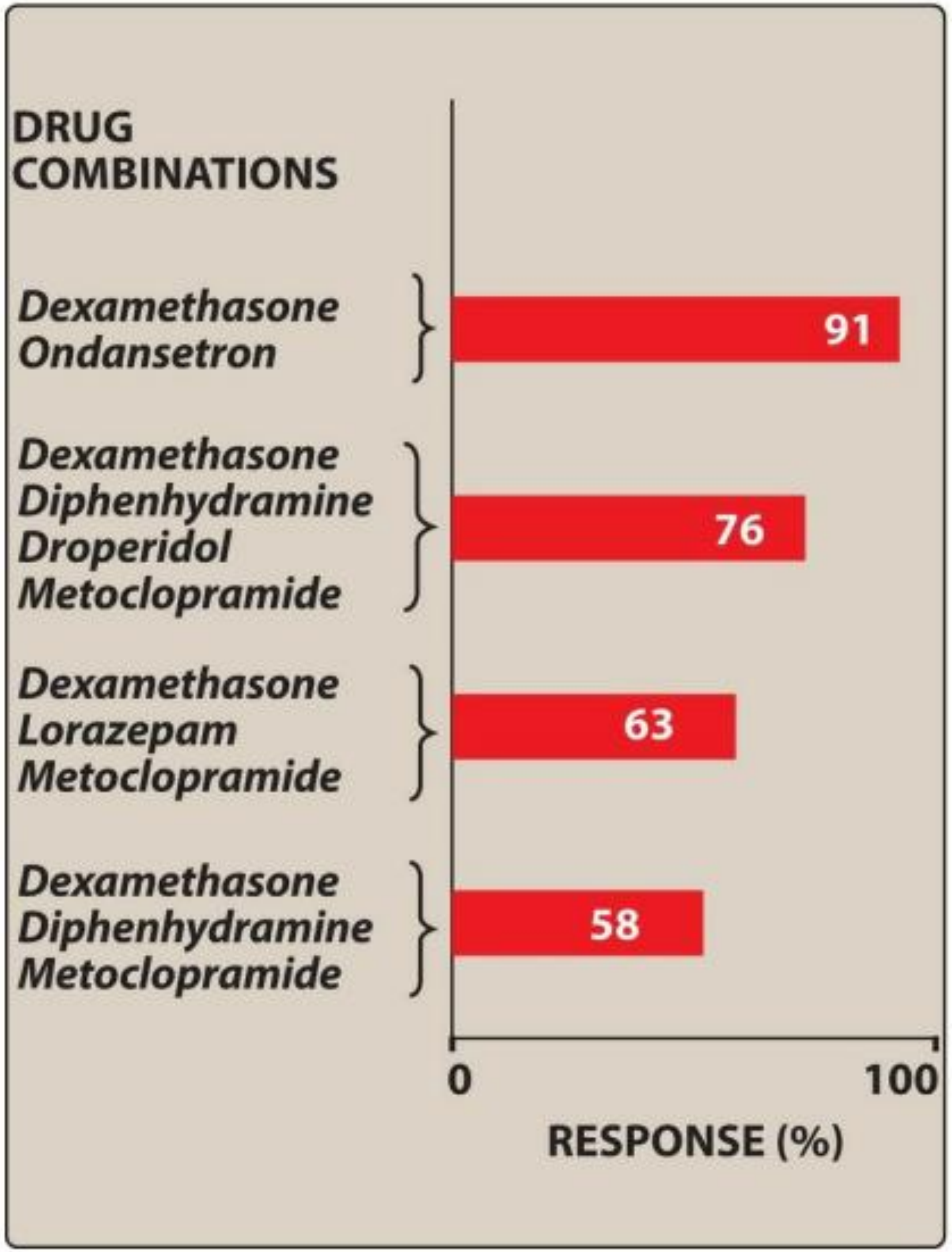
Lorazepam

Combination regimens

Antiemetic drugs are often combined to increase efficacy or decrease toxicity. Corticosteroids, most commonly **dexamethasone**, increase antiemetic activity when given with high-dose **metoclopramide, a 5-HT3 antagonist, phenothiazine, butyrophenone, or a benzodiazepine.**

Antihistamines, such as diphenhydramine, are often administered in combination with **high-dose metoclopramide** to reduce extrapyramidal reactions or with **corticosteroids** to counter metoclopramide-induced diarrhea. Addition of a substance P/neurokinin-1 receptor antagonist to a **5-HT3 antagonist and dexamethasone** is beneficial in highly emetogenic regimens, especially those with delayed CINV (chemotherapy induced nausea and vomiting)

Effectiveness of antiemetic activity of some drug combinations against emetic episodes in the first 24 hours after cisplatin chemotherapy



Clinical uses of antiemetic drugs

- Histamine H_1 receptor antagonists
  - **Cyclizine**: motion sickness, vestibular disorders, nausea and vomiting associated with surgery and postoperative narcotic analgesic use.
  - **Cinnarizine**: motion sickness, vestibular disorders (e.g. Menière's disease).
  - **Promethazine**: severe morning sickness of pregnancy, motion sickness, vestibular disorders.
- Muscarinic receptor antagonists:
  - **Hyoscine**: motion sickness.
- Dopamine D_2 receptor antagonists:
  - Phenothiazines (e.g. **prochlorperazine**): vomiting caused by migraine, vestibular disorders, radiation, viral gastroenteritis, severe morning sickness of pregnancy.
  - **Metoclopramide**: vomiting caused by migraine, radiation, gastrointestinal disorders, cytotoxic drugs, prevention of nausea and vomiting in the postoperative period.
  - **Domperidone** is less liable to cause central nervous system side effects in patients with Parkinson's disease as it penetrates the blood–brain barrier poorly.
- 5-Hydroxytryptamine $(5\text{-HT})_3$ receptor antagonists (e.g. **ondansetron**): cytotoxic drugs or radiation, postoperative vomiting.
- Cannabinoids (e.g. **nabilone**): cytotoxic drugs
- NK_1 receptor antagonists (e.g. fosaprepitant): cytotoxic

Table 31.1 Sites of action of common antiemetic drugs

Class	Drugs	Site of action	Comments
Antihistamines	Cinnarizine, cyclizine, promethazine	H_1 receptors n the CNS (causing sedation) and possibly anticholinergic actions in the vestibular apparatus	Widely effective regardless of cause of emesis
Antimuscarinics	Hyoscine	Anticholinergic actions in the vestibular apparatus and possibly elsewhere	Mainly motion sickness
Cannabinoids	Nabilone	Probably CB_1 receptors in the GI tract	CINV in patients where other drugs have been ineffective
Dopamine antagonists	Phenothiazines: prochlorphenazine, perphenazine, trifluorphenazine, chlorpromazine	D_2 receptors in CTZ	CINV, PONV, RS
	Related drugs: droperidol, haloperidol	D_2 receptors in GI tract	CINV, PONV, RS
	Metoclopramide	D_2 receptors in the CTZ and GI tract	PONV, CINV
Glucocorticoids	Dexamethasone	Probably multiple sites of action, including the GI tract	PONV, CINV; typically used in combination with other drugs
$5\text{-}HT_3$ antagonists	Granisteron, ondansetron, palonosetron	$5\text{-}HT_3$ receptors in CTZ and GI tract	PONV, CINV
Neurokinin-1 antagonists	Aprepitant, fosaprepitant	NK_1 receptors in CTZ, vomiting centre and possibly the GI tract	CINV; given in combination with another drug

5-HT, 5-hydroxytryptamine; *CINV,* cytotoxic drug-induced vomiting; *CNS,* central nervous system; *CTZ,* chemoreceptor trigger zone; *GI,* gastrointestinal; *PONV,* postoperative nausea and vomiting; *RS,* radiation sickness.

The reflex mechanism of vomiting

Emetic stimuli include:

- chemicals or drugs in the blood or intestine;
- neuronal input from the gastrointestinal (GI) tract, labyrinth and central nervous system (CNS).

Pathways and mediators include:

- impulses from the chemoreceptor trigger zone and various other CNS centres relayed to the vomiting centre;
- chemical transmitters such as histamine acetylcholine, dopamine, 5-hydroxytryptamine (5-HT) and substance P, acting on H_1, muscarinic, D_2, 5 HT_3 and NK_1 receptors, respectively.

Antiemetic drugs include:

- H_1 receptor antagonists (e.g. **cinnarizine**);
- muscarinic antagonists (e.g. **hyoscine**);
- 5-HT_3 receptor antagonists (e.g. **ondansetron**);
- D_2 receptor antagonists (e.g. **metoclopramide**);
- cannabinoids (e.g. **nabilone**);
- neurokinin-1 antagonists (e.g. **aprepitant**, **fosaprepitant**).

Main side effects of principal antiemetics include:

- drowsiness and antiparasympathetic effects (**hyoscine**, **nabilone** > **cinnarizine**);
- dystonic reactions (**metoclopramide**);
- general CNS disturbances (**nabilone**);
- headache, GI tract upsets (**ondansetron**).

REFERENCES

- Lippincott® Illustrated Reviews: Pharmacology 7TH Edition 2019. Chapter 40, Gastrointestinal and Antiemetic Drugs, Carol Motycka and Adonice Khoury. pages 1552-1559.

Questions